

CSCI E-106:Assignment 1

Due Date: February 6, 2026 at 11:59 pm EST

Instructions

Students should submit their reports on Canvas. The report needs to clearly state what question is being solved, step-by-step walk-through solutions, and final answers clearly indicated. Please solve by hand where appropriate.

Please submit two files: (1) a R Markdown file (.Rmd extension) and (2) a PDF document, word, or html generated using knitr for the .Rmd file submitted in (1) where appropriate. Please, use RStudio Cloud for your solutions.

Problem 1

The dataset prostate comes from a study on 97 men with prostate cancer who were due to receive a radical prostatectomy. You can download this data set by installing faraway library. To get the data set, copy and paste the r command: `install.packages("faraway"); library(faraway); data(prostate, package="faraway")`. (100 points)

The response variable for this assignment is `lpsa` (log of prostate-specific antigen). The dataset includes the following variables:

- `lcavol`: log(cancer volume)
- `lweight`: log(prostate weight)
- `age`: age of patient
- `lbph`: log(benign prostatic hyperplasia amount)
- `svi`: seminal vesicle invasion (indicator)
- `lcp`: log(capsular penetration)
- `gleason`: Gleason score
- `pgg45`: percentage Gleason scores 4 or 5
- `lpsa`: log(prostate specific antigen)

- a-) How many observations are in the dataset? (10 points)
- b-) Calculate the mean and median of each variable. Discuss whether any variables appear to have outliers, and briefly describe how you identified them. (20 points)
- c-) Compute the correlation matrix for all variables. Create scatterplots of each predictor versus the response variable `lpsa`. Identify which predictors appear to be strongly correlated with `lpsa`, using both the correlations and the plots to justify your answer. (20 points)
- d-) Build separate simple linear regression models, by using `lcavol`, `lweight` and `age` to explain `lpsa`. For each model, clearly state the fitted regression equation with estimated coefficients. (25 points)
- e-) For each regression model from part (d):
 - Plot `lpsa` versus each predictor separately.

- Add the fitted regression line to each plot.
- Based on the models and plots, choose the best single-predictor model and explain your reasoning(25 points)